

Name : Yaseen Hamdy

1-create two workspaces dev and prod

```
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace new dev
```

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state, so if you run "terraform plan" Terraform will not see any existing state for this configuration.

```
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace list
```

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

```
default
* dev
```

```
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace new prod
```

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state, so if you run "terraform plan" Terraform will not see any existing state for this configuration.

```
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace list
```

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

```
default
dev
* prod
```

2-create two variable definition files(.tfvars) for the two environments

```
dev.tfvars > abc aws_region  
1 aws_region = "eu-central-1"
```

```
prod.tfvars > abc aws_region  
1 aws_region = "us-east-2"
```

```
variables.tf > 🔗 variable "aws_region"  
1 variable "aws_region" {  
2     type = string  
3     default = "us-east-1"  
4 }
```

3-separate network resources into network module

```
main.tf > module "network"
1  module "network" {
2      source = "../network/"
3
4  }
```

outputs of module

```
network > outputs.tf > output "public_1_subnet"
1  output "vpc" {
2      value = aws_vpc.main
3  }
4  output "public_1_subnet" {
5
6      value = aws_subnet.public_1
7  }
8
9  output "public_2_subnet" {
10
11      value = aws_subnet.public_2
12  }
13
14  output "private_1_subnet" {
15
16      value = aws_subnet.private_1
17  }
18
19  output "private_2_subnet" {
20
21      value = aws_subnet.private_2
22  }
23
24
```

4-apply your code to create two environments one in us-east-1 and eu-central-1

make resources in 3 regions

- default → us-east-1
- dev → eu-central-1
- prod → us-east-2

Instances (2) Info

Last updated 15 minutes ago

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type
application	i-03d81b5192f1beaf7	Running	t3.micro
bastion	i-063aa905819c80078	Running	t3.micro

Ohious-east-2

N. Californiaus-west-1

Oregonus-west-2

Asia Pacific

Mumbaiap-south-1

Osakaap-northeast-3

Seoulap-northeast-2

Singaporeap-southeast-1

Launch instances

< 1 >

Availability Zone

us-east-2a

us-east-2a

Instances (4) Info

Last updated less than a minute ago

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type
application	i-03b06cad871e31167	Terminated	t3.micro
bastion	i-0bc17782f75649874	Terminated	t3.micro
application	i-060637a5db4b21c04	Running	t3.micro
bastion	i-037c4466f31c8d13a	Running	t3.micro

N. Virginiaus-east-1

Ohious-east-2

N. Californiaus-west-1

Oregonus-west-2

Asia Pacific

Mumbaiap-south-1

Osakaap-northeast-3

Seoulap-northeast-2

Singaporeap-southeast-1

Sydneyap-southeast-2

Tokyop-northeast-1

Launch

Availability Zone

us-east-2a

us-east-2a

Instances (2) Info

Last updated less than a minute ago

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type
bastion	i-0f9ace1322d3ccd7c	Running	t3.micro
application	i-06ad27fb92f19ac21	Running	t3.micro

Seoulap-northeast-2

Singaporeap-southeast-1

Sydneyap-southeast-2

Tokyop-northeast-1

Canada

Centralca-central-1

Europe

Frankfurt eu-central-1

Ireland eu-west-1

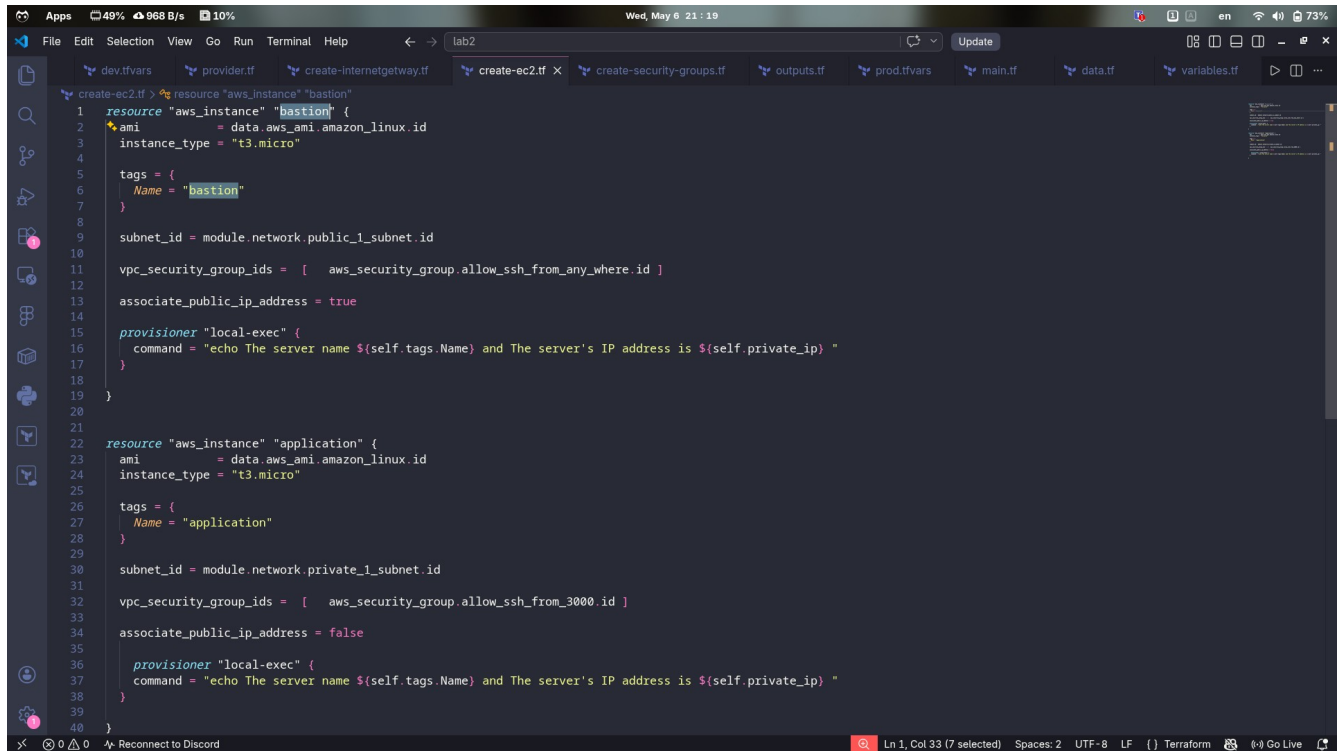
Launch

Availability Zone

us-east-2a

us-east-2a

5-run local-exec provisioner to print the public_ip of bastion ec2



The screenshot shows a VS Code editor with several Terraform configuration files open. The active file is `create-ec2.tf`, which defines two AWS instances: `bastion` and `application`. Both instances use the `local-exec` provisioner to print their names and private IP addresses. The `bastion` instance is associated with the `allow_ssh_from_anywhere` security group and has `associate_public_ip_address = true`. The `application` instance is associated with the `allow_ssh_from_3000` security group and has `associate_public_ip_address = false`.

```
1 resource "aws_instance" "bastion" {
2   ami           = data.aws_ami.amazon_linux.id
3   instance_type = "t3.micro"
4
5   tags = {
6     Name = "bastion"
7   }
8
9   subnet_id = module.network.public_1_subnet.id
10
11   vpc_security_group_ids = [ aws_security_group.allow_ssh_from_anywhere.id ]
12
13   associate_public_ip_address = true
14
15   provisioner "local-exec" {
16     command = "echo The server name ${self.tags.Name} and The server's IP address is ${self.private_ip} "
17   }
18 }
19
20
21 resource "aws_instance" "application" {
22   ami           = data.aws_ami.amazon_linux.id
23   instance_type = "t3.micro"
24
25   tags = {
26     Name = "application"
27   }
28
29   subnet_id = module.network.private_1_subnet.id
30
31   vpc_security_group_ids = [ aws_security_group.allow_ssh_from_3000.id ]
32
33   associate_public_ip_address = false
34
35   provisioner "local-exec" {
36     command = "echo The server name ${self.tags.Name} and The server's IP address is ${self.private_ip} "
37   }
38 }
39
40 }
```

```
yaseen@fedora:~/iti-track/Terraform/lab2$ terraform taint aws_instance.bastion
Warning: Deprecated Parameter
The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Acquiring state lock. This may take a few moments...

Warning: Deprecated Parameter
The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Resource instance aws_instance.bastion has been marked as tainted.
yaseen@fedora:~/iti-track/Terraform/lab2$ terraform taint aws_instance.application
Warning: Deprecated Parameter
The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Acquiring state lock. This may take a few moments...

Warning: Deprecated Parameter
The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

Resource instance aws_instance.application has been marked as tainted.
yaseen@fedora:~/iti-track/Terraform/lab2$
```

```
aws_instance.application: Provisioning with 'local-exec'...
aws_instance.application (local-exec): Executing: ["/bin/sh" "-c" "echo The server name application and The server IP address is 10.0.3.83 "]
aws_instance.application (local-exec): The server name application and The server IP address is 10.0.3.83
aws_instance.application: Creation complete after 15s [id=i-09241ffc146d27d41]
aws_instance.bastion: Still creating... [00m10s elapsed]
aws_instance.bastion: Provisioning with 'local-exec'...
aws_instance.bastion (local-exec): Executing: ["/bin/sh" "-c" "echo The server name bastion and The server IP address is 10.0.1.120 "]
aws_instance.bastion (local-exec): The server name bastion and The server IP address is 10.0.1.120
aws_instance.bastion: Creation complete after 14s [id=i-0ab00bf63fa25ffe0]
```

6- upload infrastructure code on github project

repo link : <https://github.com/yaseenhamdy/terraform-labs.git>

 yaseenhamdy terraform labs2	b3acbbd · now	
 network	terraform labs2	now
 .gitignore	terraform labs2	now
 .terraform.lock.hcl	terraform labs2	now
 .~lock.lab2.odt#	terraform labs2	now
 create-ec2.tf	terraform labs2	now
 create-security-groups.tf	terraform labs2	now
 data.tf	terraform labs2	now
 lab2.odt	terraform labs2	now
 main.tf	terraform labs2	now
 provider.tf	terraform labs2	now
 state.tf	terraform labs2	now
 variables.tf	terraform labs2	now

7-verify your email in ses service

```
create-sees.tf > ...  
1  resource "aws_ses_email_identity" "email_1" {  
2    email = "yaseenhamdy74@gmail.com"  
3  }  
4  
5  resource "aws_ses_email_identity" "email_2" {  
6    email = "yaseenhamdy888@gmail.com"  
7  }  
8
```


8- create lambda function to send email

→ IAM policy

```
modules > lambda > IAM_policy.tf > resource "aws_iam_policy" "lambda_ses_policy"
1
2  resource "aws_iam_policy" "lambda_ses_policy" {
3      name           = "lambda_ses_policy"
4      path           = "/"
5      description    = "IAM policy for send emails from Lambda"
6
7      policy = jsonencode({
8          Version = "2012-10-17"
9          Statement = [
10             {
11                 Effect = "Allow"
12                 Action = [
13                     "ses:SendEmail",
14                     "ses:SendRawEmail"
15                 ]
16                 Resource = "*"
17             }
18         ]
19     })
20 }
21
```

→ IAM Role

```
modules > lambda > IAM_role.tf > ...
1  resource "aws_iam_role" "lambda_role" {
2      name = "lambda_role"
3
4      assume_role_policy = jsonencode({
5          Version = "2012-10-17"
6          Statement = [
7              {
8                  Action = "sts:AssumeRole"
9                  Effect = "Allow"
10                 Principal = {
11                     Service = "lambda.amazonaws.com"
12                 }
13             }
14         ]
15     })
16 }
17
```

→ attach policy in role

```
modules > lambda > role_policy_attach.tf > ...
1  resource "aws_iam_role_policy_attachment" "lambda_role_policy" {
2      role          = aws_iam_role.lambda_role.name
3      policy_arn    = aws_iam_policy.lambda_ses_policy.arn
4  }
```

→ archiving code file

```
modules > lambda > data-lambda-archiving.tf > data "archive_file" "lambda_archiving"
1  data "archive_file" "lambda_archiving" {
2      type          = "zip"
3      source_file   = "${path.module}/send_email.py"
4      output_path   = "${path.module}/send_email.zip"
5  }
```

→ create lambda function

```
modules > lambda >  lambda-func.tf >  resource "aws_lambda_function" "example"

1  resource "aws_lambda_function" "example" {
2      filename      = data.archive_file.lambda_archiving.output_path
3      function_name = "ses_lambda_function"
4      role           = aws_iam_role.lambda_role.arn
5      handler        = "send_email.lambda_handler"
6      code_sha256     = data.archive_file.lambda_archiving.output_base64sha256
7
8      runtime = "python3.12"
9
10 }
```

ses_lambda_function

[Throttle](#) [Copy ARN](#) [Actions](#)

▼ Function overview [Info](#)

[Diagram](#) [Template](#)

 **ses_lambda_function**

 Layers (0)

[+ Add trigger](#) [+ Add destination](#)

Function ARN
 `arn:aws:lambda:us-east-1:171558409860:function:ses_lambda_function`

Description
-

Last modified
9 minutes ago

[Export to Infrastructure Composer](#) [Download](#)

[Code](#) [Test](#) [Monitor](#) [Configuration](#) [Aliases](#) [Versions](#)

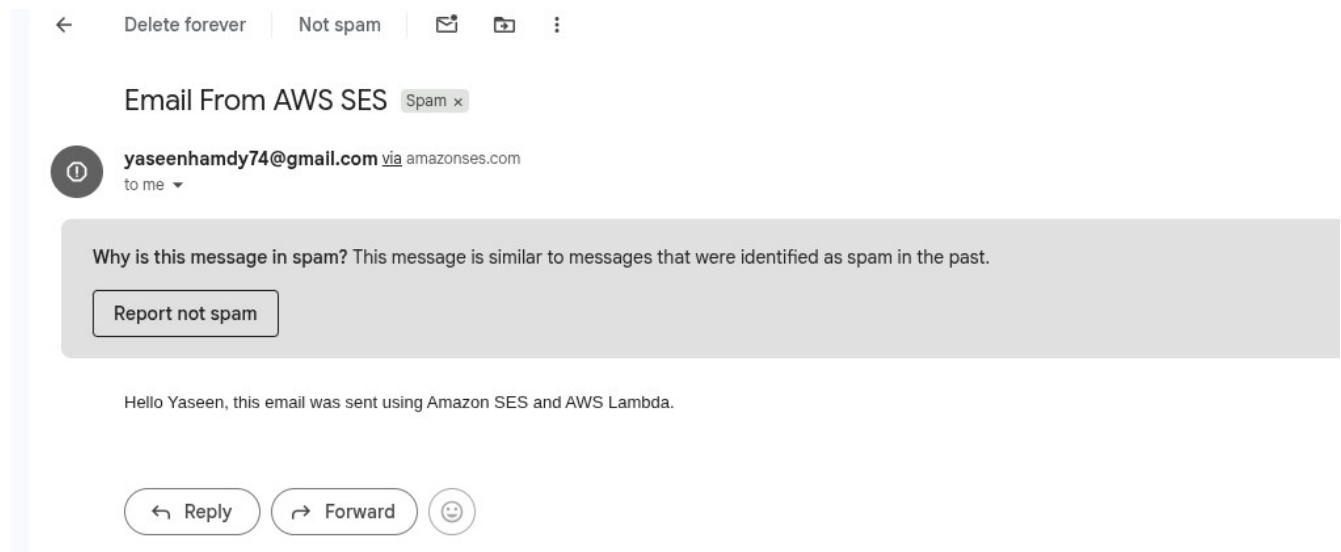
Code source [Info](#)

[Open in Visual Studio Code](#) [Update](#)

← →



9-create trigger to detect changes in state file and send the email



10-create rds (mysql)

```
modules > rds > create-sql-rds.tf > resource "aws_db_instance" "sqlDB"
1  resource "aws_db_instance" "sqlDB" {
2      allocated_storage      = 10
3      db_name                 = var.db_name
4      engine                  = "mysql"
5      engine_version          = "8.0"
6      instance_class          = "db.t3.micro"
7      username                = var.db_username
8      password                = var.db_password
9      parameter_group_name    = "default.mysql8.0"
10     skip_final_snapshot     = true
11 }
```

terraform-20260507172058178600000001

  [Modify](#) [Actions](#) ▼

Summary

DB identifier
terraform-
202605071720581786000
00001

CPU
 5.07%

Status
 Available

Class
db.t3.micro

Role
Instance

Current activity
 0 Connections

Engine
MySQL Community

Region & AZ
us-east-1c

Recommendations

< [Connectivity & security](#) [Monitoring](#) [Logs & events](#) [Configuration](#) [Zero-ETL integrations](#) [Maintenance & backup](#) >

11-create elastic cache (redis)

```
modules > rds >  create-sql-rds.tf > ...  
1  resource "aws_db_instance" "sqlDB" {  
2      allocated_storage      = 10  
3      db_name                 = var.db_name  
4      engine                 = "mysql"  
5      engine_version         = "8.0"  
6      instance_class         = "db.t3.micro"  
7      username               = var.db_username  
8      password               = var.db_password  
9      parameter_group_name   = "default.mysql8.0"  
10     skip_final_snapshot    = true  
11 }  
12  
13
```

app-redis [Info](#)

[Upgrade to Valkey](#)[Modify](#)[Act](#)

▼ Cluster details

Cluster name app-redis	Description -	Node type cache.m4.large	Status  Available
Engine Redis	Engine version 7.1.0	Global datastore -	Global datastore role -
Update status Up to date	Cluster mode Disabled	Shards 0	Number of nodes 1
Data tiering Disabled	Multi-AZ Disabled	Auto-failover Disabled	Encryption in transit Disabled
Encryption at rest Disabled	Parameter group default.redis7	Outpost ARN -	Configuration endpoint -
Primary endpoint -	Reader endpoint -	ARN  arn:aws:elasticache:us-east-1:171558409860:cluster:app-redis	Data migration No active migrations